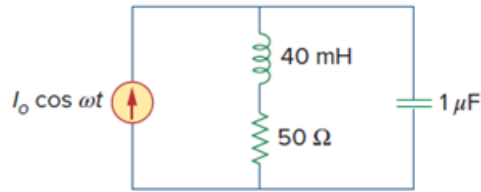


Problem

For the “tank” circuit of Fig. 14.79, obtain the frequency response (voltage across the capacitor) using *PSpice* or *MultiSim*. Determine the resonant frequency of the circuit.

Figure 14.79:



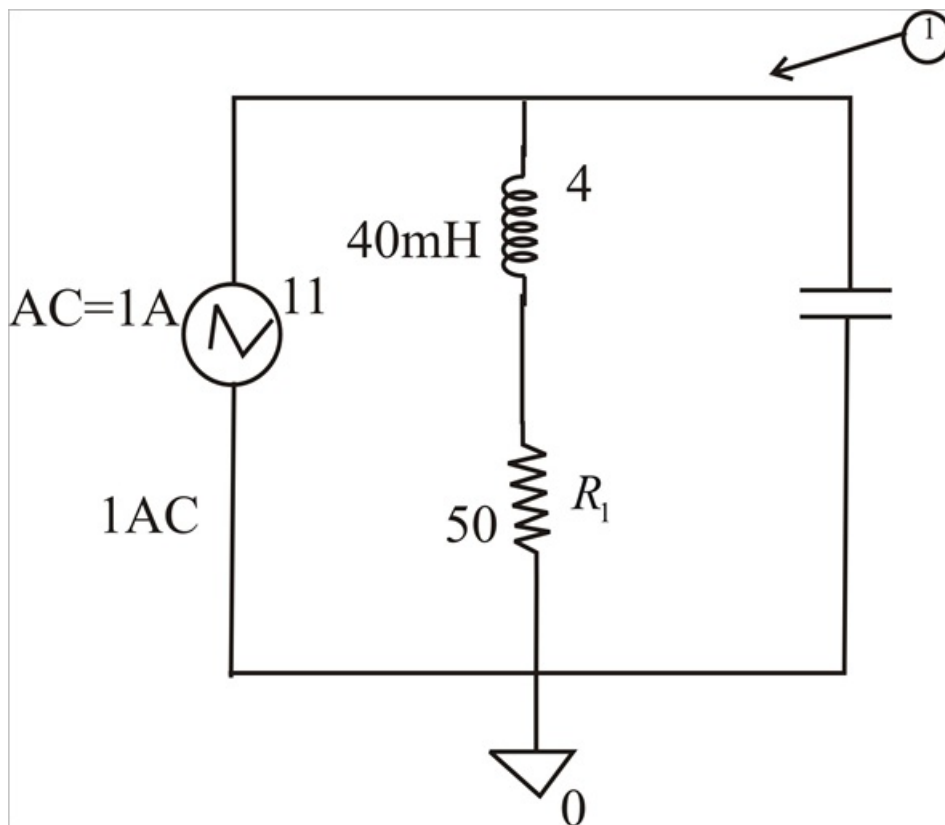
Step-by-step solution

Step 1 of 3

The schematic is shown below. In the A.C sweep box, we select total points = 100, start frequency = 10, and end frequency = 10 K. After simulation, the magnitude plot of the frequency response is obtained from the plot we obtain the resonant frequency F_o is approximately equal to 800 Hz. So that

$$\omega_o = 2\pi f_o = \underline{5026 \text{ rad/s}}$$

Step 2 of 3



Step 3 of 3

